Title: Machine-Learning Sepsis Early Warning Systems in Adult ICUs: A Synthetic PRISMA 2020 Systematic Review

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Background: Automated sepsis alerts aim to shorten time-to-antibiotics in adult intensive care units, yet prospective clinical impact remains heterogeneous.

Objectives: Synthesize prospective ICU implementations evaluating mortality, antibiotic timing, and false-alarm burden under PRISMA 2020 guidelines.

Methods: MEDLINE, Embase, Cochrane CENTRAL, and IEEE Xplore were searched through June 2026 (PROSPERO CRD42026000999-SYN). Risk of bias was assessed via ROBINS-I and RoB 2.

Results: Of 3,420 records identified (610 duplicates removed), 2,810 titles/abstracts were screened, 2,664 excluded, 146 full texts assessed, 128 excluded, and 18 prospective studies (N = 94,300 ICU admissions; 6 RCTs, 12 interv

Waveform+EHR multimodal models achieved median AUROC 0.87 (IQR 0.84-0.90) vs 0.78 (0.74-0.81) for vitals-only alerts, reducing median time-to-antimicrobials by 1.8 hours.

Limitations: High alert fatigue in uncalibrated thresholds; heterogeneous Sepsis-3 adjudication across sites.

Conclusions: Prospective ICU sepsis warning systems consistently accelerate antimicrobial delivery, while mortality benefit clusters in studies pairing human nurse-physician workflow integration.